05-24 第一篇

The Social Structure of Ants: A Tiny but Complex World

Ants are among the most successful creatures on Earth. There are over 12,000 known species, and they live on every continent except Antarctica. What makes them so successful? [1] In fact, ants work together in colonies（蚁群）—each ant has a role, and the group achieves things that a single ant never could. A colony can build huge nests, find food over long distances, and defend itself against enemies. That is why ants are often called "social insects." Their strength lies in cooperation, not in size or speed. Together, they form one of the most organised societies in the animal kingdom.

How is a colony organised? [2] Most colonies have a queen—she lays all the eggs and is the mother of the colony. Worker ants are females that do not reproduce; they gather food, care for the young, and build the nest. Some species also have soldier ants that defend the colony. Male ants usually appear only when it is time to mate, and they die soon after. In other words, each ant has a clear job, and the colony works like a single super-organism. No single ant knows the whole plan, but together they achieve amazing things.

How do ants communicate? [3] Ants use chemicals called pheromones（信息素）to send messages. When a worker finds food, she leaves a trail of pheromones so others can follow. Ants also touch their antennae（触角）to share information. They can tell nest-mates from strangers by smell. For this reason, an ant colony is like a living network—every ant is connected, and information flows quickly through the group. This allows thousands of ants to act as one, even without a central "boss."

Why do ants matter to us? [4] Ants help ecosystems in many ways. They spread seeds, turn over soil, and control pest populations. Some ants even work as "farmers," growing fungus（真菌）for food. Scientists study ants to learn about cooperation, networks, and problem-solving. It is true that we can learn a lot from these tiny creatures. Their colonies show us how simple rules can lead to complex, organised behaviour. Understanding ants helps us understand nature—and maybe even ourselves.

[5] In short, ants are a model of teamwork and efficiency. Each ant is small, but together they build, defend, and explore. From the rainforest to your garden, ant colonies are busy at work every day. Remember: the next time you see a line of ants carrying food, you are watching millions of years of evolution in action. So take a moment to watch—these tiny insects have lessons for us all about working together and sharing the load.

选项

A. That is because ants succeed through teamwork and division of labour. B. In other words, the colony is divided into roles that work as one. C. That is why communication is key to ant society. D. It is true that we can learn a lot from these tiny creatures. E. In conclusion, ants are a powerful example of teamwork in nature. F. For example, some ant species build nests under tree roots.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：总领蚂蚁通过合作取得成功。空前描述"Ants are among the most successful creatures on Earth...What makes them so successful?"，空后说明"In fact, ants work together in colonies—each ant has a role, and the group achieves things that a single ant never could...ants are often called 'social insects'...Their strength lies in cooperation, not in size or speed"。A选项"That is because ants succeed through teamwork and division of labour"（那是因为蚂蚁通过团队合作和分工获得成功）中"That is because"解释蚂蚁成功之道，"teamwork and division of labour"与空后work together、each ant has a role、cooperation精准对应，衔接顺畅。
- **[2] B**：
  - **线索**：蚁群内的角色分工。空前设问"How is a colony organised?"，空后列举"Most colonies have a queen—she lays all the eggs...Worker ants are females...they gather food, care for the young, and build the nest...Some species also have soldier ants...Male ants usually appear only when it is time to mate...each ant has a clear job, and the colony works like a single super-organism"。B选项"In other words, the colony is divided into roles that work as one"（换言之，蚁群被划分为各司其职却又协同运作的角色）中"In other words"概括空后内容，"divided into roles that work as one"与空后queen、workers、soldiers、single super-organism完美呼应，衔接自然。
- **[3] C**：
  - **线索**：蚂蚁的沟通方式与信息传递。空前设问"How do ants communicate?"，空后详述"Ants use chemicals called pheromones to send messages...When a worker finds food, she leaves a trail of pheromones so others can follow...Ants also touch their antennae to share information...an ant colony is like a living network—every ant is connected...This allows thousands of ants to act as one"。C选项"That is why communication is key to ant society"（因此沟通是蚂蚁社会的关键）中"That is why"承接空前及空后结论，"communication is key"与

空后pheromones、antennae、living network、information flows精准对应，逻辑紧密。

- [4] D：
  - **线索**：蚂蚁对人类与生态的意义及可借鉴之处。空前设问"Why do ants matter to us?"，空后说明"Ants help ecosystems in many ways...They spread seeds, turn over soil, and control pest populations...Scientists study ants to learn about cooperation, networks, and problem-solving...we can learn a lot from these tiny creatures...Their colonies show us how simple rules can lead to complex, organised behaviour"。D选项"It is true that we can learn a lot from these tiny creatures"（的确，我们可以从这些小小生物身上学到很多）中"It is true that"加强衔接，"learn a lot from these tiny creatures"与空后study ants、learn about cooperation、understand nature完美呼应，衔接得当。
- [5] E：
  - **线索**：全文收束与蚂蚁的团队精神启示。空[5]位于结尾段段首，空后总结"ants are a model of teamwork and efficiency...Each ant is small, but together they build, defend, and explore...the next time you see a line of ants carrying food, you are watching millions of years of evolution in action...these tiny insects have lessons for us all about working together and sharing the load"。E选项"In conclusion, ants are a powerful example of teamwork in nature"（总之，蚂蚁是自然界团队合作的鲜明例证）用"In conclusion"总结全文，"powerful example of teamwork"与空后model of teamwork、lessons about working together形成呼应，收束有力。
- **干扰项分析**：
  - **F. For example, some ant species build nests under tree roots.**：涉及蚂蚁在树根下筑巢，易被误填[2]或[4]。但F讨论的是筑巢地点的具体案例，而空[2]聚焦蚁群内的角色分工（queen、workers、soldiers），空[4]强调蚂蚁对生态与人类的启示；F属于单一现象举例，与空[2]的社会结构、空[4]的学习价值论述层级不同。F属"主题相关但论述层级错位"型干扰。